

sPellorama: An Immersive Prototyping Tool using Generative Panorama and Voice-to-Prompts

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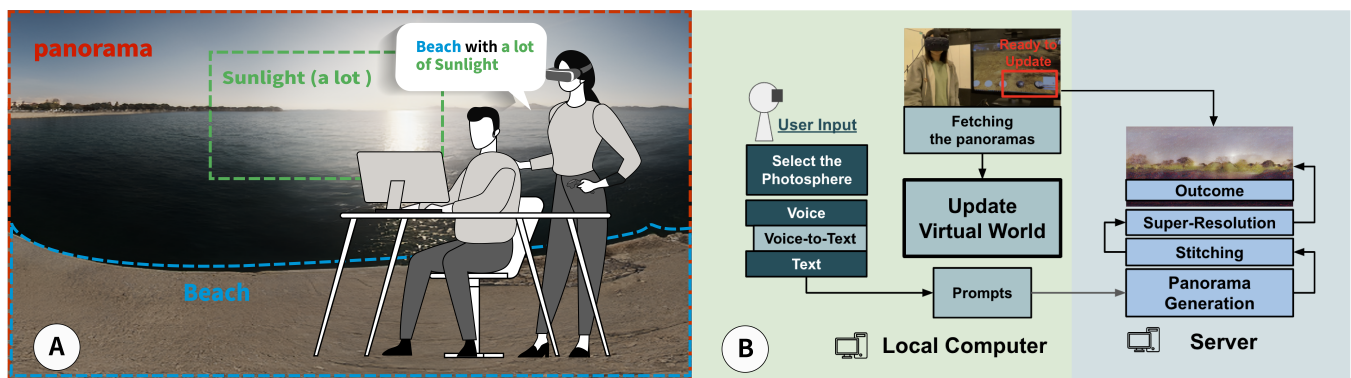


Figure 1: Overview: We proposed an immersive tool that enables Virtual Reality (VR) content creators to quickly prototype scenes based on verbal input. A: Assumed Scenario. B: Overall Workflow

ABSTRACT

We proposed sPellorama, an immersive tool that enables Virtual Reality (VR) content creators to quickly prototype scenes based on verbal input. The system first converts voice input to text, then utilizes a text-guided panorama generation model to produce the described scene. The panorama is later applied to Skybox in Unity. Previously generated panorama will be preserved in the photosphere and ready to be viewed. The pilot study shows that our tool can enhance the process of discussion and prototyping for VR content creators.

CCS CONCEPTS

• Computing methodologies → Virtual reality; • Human-centered computing → Interaction design process and methods.

KEYWORDS

Virtual Reality, VR Prototyping, Generative Panorama

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1 INTRODUCTION

In an immersive story or 360 video production, the environment setting of the story is crucial as it will determine how to place the characters, props and sets in the scenario. The traditional way of planning for the environment setting requires directors to do immersive storyboarding. One of the techniques is to sketch an existing flat photosphere image, then fold it up for viewing or scan the photo and view it in a photosphere viewer or game engine.

To speed up such processes, recent researches help Virtual Reality (VR) content creators visualize their ideas. 360proto[Nebeling and Madier 2019] utilizes paper mock-ups to create a virtual environment by taking a photo of 360 storyboard sketches and previewing it in Google Cardboard. [Speicher et al. 2021] propose a medium-fi prototyping method for Augmented Reality(AR) and VR using

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360 theaters, aiming to help designers with a non-programming background to produce prototypes with higher fidelity.[Artizzu et al. 2022]propose a configurable virtual reality template that consists of VR-ready 3D models and scenes for end users to create VR environments. These approaches can help VR content creators with a non-programming background to deliver their ideas quickly for discussion. However, some of these approaches still require sketching or are only available to premade virtual environments. Recent advancements in generative AI have shown another possibility to address this issue. Instead of generating the scene on a full 3D basis, the generation of panoramic photos can leverage the result of image-generating techniques while still capable of being used as a scene in virtual reality. With the development of various models capable of generating photo-realistic images from natural language inputs[Ramesh et al. 2021], coarse scenes can be generated using only natural language as input.

In this work, we proposed an immersive tool that utilizes a text-guided panorama-generating model to generate images that can be used as skybox material in the Unity game engine. This tool takes users' voice input as a prompt, passes it to the server for image generation, and then returns the result to VR as a prototype scene. Our tool lets users conveniently switch through different scenes, generating them with verbal descriptions all within the virtual environment.

2 SYSTEM OVERVIEW

We use Unity as the game engine to implement our concept of using the generated panorama to create a virtual environment. Our user interface consists of an input field and two buttons for generating and applying a panorama. First, we ask VR content creators to input their prompts by typing or speaking. Then, we use Unity's web request to send the text prompt.

After receiving input, the server utilizes Text2Light[Chen et al. 2022] text-driven panoramic generation process to produce a low-resolution 360-degree photograph based on the input. It first generates a holistic vector based on text input, and the vector is then used to guide the addition of finer details through a local sampler concerning spherical coordinates. Subsequently, the resulting image is obtained through decoding from the local sampler. Afterward, the image edges are carefully stitched and blended with adjacent pixels, ensuring a natural appearance upon projection onto a spherical surface. Finally, the resulting panorama is super-resolution for a VR skybox.

After the generation process is finished, the result is fetched by Unity from the server. In our user interface, VR content creators can directly use the panorama as a skybox by selecting the corresponding photosphere. The photospheres with textures are ready to be updated (figure 1b) while the white one still waits to fetch panorama from the server. After applying the panorama as the skybox, they can view the scene using VR Headsets. If they are unsatisfied with the outcome, they can input a new prompt to get another iteration of the panorama as a skybox to aid them in their prototyping session. The panorama that was generated previously will be saved in the photosphere accordingly, and this can be used in the discussion session when the panorama from the new prompt is still being generated.

3 PILOT STUDY

Our pilot study is conducted among 4 users with VR design experience. Each user is asked to generate at least two scenes. Then they are questioned whether this tool can help them convey their idea and how it can change their process of designing VR experiences.

All of our users agree that this tool can assist them in conveying their idea in ways such as providing visualization for discussion and validating their idea. In addition, users mention that this is a potential tool for VR pre-production with its ability to preserve human resources for making concept art. In the interview, we utilized the questions from System Usability Scale [Brooke 1995] to explore our tool's usability. User report that when given prompts that are too complicated, or include too specific details, the model will fail to follow the text description. In addition, even though we employed a state-of-the-art model, the result is far too coarse to use in production. To further enhance the usability of this tool, pre-parsing of prompts is required. Furthermore, a user also mentions that generating multiple candidate images for each prompt so could provide more choices that would satisfy the needs of the users.

4 DISCUSSION AND FUTURE WORKS

We explore the potential application of natural-language-guided panorama generation in VR prototyping and production. It allows content creators to conveniently go through various potential scene settings immersively, without the effort to construct the 3d scene itself. Instead, a coarse scene can be generated solely by a short description, which is more time and cost-efficient.

For future work, our goal is to enhance the quality of the generated images and improve the usability of the results, even in a production environment. Additionally, we plan to provide users with the ability to mark a specific region to modify their results using additional commands as a compensation method, enabling them to generate more satisfying outcomes. We also intend to conduct more comprehensive user studies to enhance the integration of the generative model into VR prototyping and design.

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